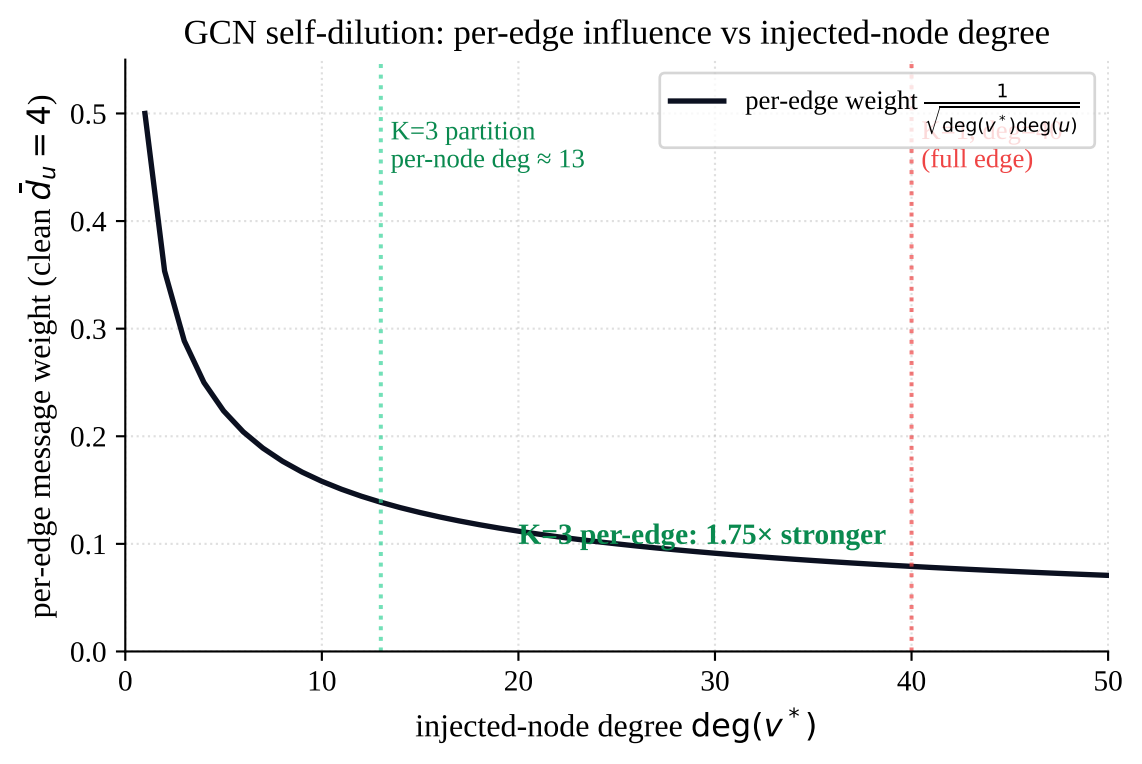
GCN self-dilution: per-edge influence vs injected-node degree
per-edge weight $\frac{1}{\sqrt{\deg(v^*)\deg(u)}}$
K=3 partition
per-node deg ≈ 13
(full edge)
K=3 per-edge: 1.75× stronger
per-edge message weight (clean $\bar{d}_u = 4$)
injected-node degree $\deg(v^*)$
0.0
0.1
0.2
0.3
0.4
0.5
0
10
20
30
40
50